Curriculum Vitae - Evan Toon - Undergraduate Physics Student

(865) 313-3875 | etoon@vols.utk.edu | evan-toon.github.io/website

Education

University of Tennessee, Expected Graduation May 2027 Knoxville, TN

Physics(BS), Academic Concentration, Astronomy Minor

- GPA: Undergraduate - 3.89

Research Experience

Dynamic Nuclear Polarization (DNP) - Nuclear Physics Research April 2025 - Present

Dr. Nadia Fomin - University of Tennessee *Knoxville, TN*

- **Computational** - Worked closely with Professor Fomin and collaborators from Jefferson Lab and Oak Ridge National Lab to further investigate how the nuclear polarization of a sample is measured via a Nuclear Magnetic Resonance (NMR) coil within a dynamically polarized scattering target. Using polarized ammonia data from Jefferson Lab's Run Group C (RGC) and the Spin Asymmetries of the Nucleon Experiment (SANE), I used Python to perform a three-Gaussian fitting analysis to better understand potential external contributions to the NMR signal that would contribute to the overall polarization measurement.
- **In the Lab** - Worked with Professor Fomin and her graduate students to recommission a DNP target setup transferred from Oak Ridge National Lab. I actively contributed to the reassembly, optimization, and operation of the system, gaining extensive hands on expertise with the dilution refrigerator, superconducting magnet, vacuum systems, supplemental microwaves and NMR electronics, and other critical DNP components.

Science Undergraduate Laboratory Internships (SULI) - DNP Research at HFIR May 2026 - August 2026

Dr. Josh Pierce - Oak Ridge National Laboratory (ORNL) *Oak Ridge, TN*

- I actively contributed to the commissioning and optimization of a new Dynamic Nuclear Polarization (DNP) system for the IMAGINE-X BRaVE protein crystallography project at the High Flux Isotope Reactor (HFIR). I worked on the design and integration of the microwave equipped sample stick with integrated NMR circuitry, as well a new cryogenic sample space equipped with a five stage pizeo-electric goniometer that houses samples of interest in any desired orientation in the required low temperature, high field environment. Along with this I gained hands on experience operating a 1 K ^{4}He evaporation refrigerator, 5 T superconducting magnet, microwave systems, and NMR instrumentation.

Presentations

Selected Talks

(1) APS Division of Nuclear Physics Fall Meeting - Undergraduate Oral Presentation Fall 2025
(2) SPS Zone 8 Meeting - Oral Presentation Spring 2026
(3) UTK Arts & Sciences Undergraduate Research Symposium (ASUReS) Spring 2026
(4) The 21st International Workshop on Polarized Sources, Targets, and Polarimetry (PSTP) Fall 2026

Selected Posters

(1) UTK Exhibition of Undergraduate Research and Creative Achievement (EURēCA) Spring 2026
(2) D.O.E. Summer Undergraduate Laboratory Internship (SULI) Poster Presentation Summer 2026
(3) APS Division of Nuclear Physics Fall Meeting - C.E.U. Poster Session Fall 2026

Relevant Courses

Classical Mechanics I [PHYS 311]; Classical Mechanics II [PHYS 312] Fall 2025, Spring 2026

Quantum Mechanics I [PHYS 411]; Quantum Mechanics II [PHYS 412] Fall 2026, Spring 2027

Electricity & Magnetism I [PHYS 431]; Electricity & Magnetism II [PHYS 432] Fall 2026, Spring 2027

Technical Proficiencies

Programming: Python (NumPy, Matplotlib, Pandas, SciPy, lmfit), Git
Software: LaTeX, Mathematica

Awards & Honors

Dean's List Fall 2023 – Spring 2026

Community & Leadership

Society of Physics Students (SPS) Fall 2023 – Present
- *Outreach/Media Coordinator* Spring 2026
- *President* Fall 2026 – Spring 2027

Astrophysics Journal Club Fall 2023 – Present
- *Member*

Astronomy Club Fall 2025 - Present
- *Member*

Phi Beta Kappa (ΦBK) Spring 2026 - Present
- *Member*

Interests

Experimental Nuclear Physics

Dynamic Nuclear Polarization (DNP)

Cryogenic Instrumentation

- *Dilution Refrigeration,* ^{4}He *Evaporation Refrigerators, etc*

Solid Polarized Scattering Targets

Quantum Chromodynamics (QCD), Probing Nucleon Structure

Electron Scattering, Neutron Scattering

Low Temperature Physics

Neutron Crystallography